

MILESTONE 6: INTEGRATED ANALYTICS SYSTEM

Research Contribution, Advanced Analytics & Decision Support

1. Executive Summary

This milestone represents the final integration of the e-commerce analytics system. The objective was to evolve the platform from a descriptive dashboard into a predictive research tool. This was achieved by implementing a novel Retention Score model, resolving complex data anomalies, and designing a human-centered interface for strategic decision-support.

2. Advanced Data Engineering

Before implementing advanced models, a rigorous data audit was conducted. A critical anomaly was identified where the 'Items Purchased' field contained negative integers, likely representing returns or data entry errors.

- Technical Resolution:
- To maintain the integrity of the total engagement metrics, an 'Items Corrected' layer was built using the Absolute Value transformation. This ensured that the system accurately reflects gross transactional volume.

DAX: Items Corrected = ABS('cleaned_customer_data'[Items Purchased])

3. Research Contribution: The Weighted Retention Model

The system's core innovation is the Weighted Retention Score. This model goes beyond historical data by predicting future customer loyalty based on a multi-criteria weighting system.

3.1 Algorithmic Logic

The algorithm synthesizes three distinct dimensions: Satisfaction Levels, Membership Tiers, and corrected purchase frequency. Weights are assigned as follows:

- High Weight (3.0): Gold Members with 'Satisfied' status.
- Medium Weight (1.5 - 2.5): Neutral sentiment or Silver members with positive feedback.
- Low Weight (0.5): Any 'Unsatisfied' customer, regardless of membership tier.

3.2 Final DAX Formula

```
Retention Score =
SWITCH(TRUE(),
    'Table'[Membership Category] = "Gold" && 'Table'[Satisfaction Level] =
    "Satisfied", 3,
    'Table'[Membership Category] = "Silver" && 'Table'[Satisfaction Level] =
    "Satisfied", 2.5,
    'Table'[Satisfaction Level] = "Neutral", 1.5,
    'Table'[Satisfaction Level] = "Unsatisfied", 0.5,
    1
)
```

4. Integrated System Visualization

The Power BI dashboard was redesigned to serve as an Interactive Decision-Support System (DSS).

- 1. Strategic Scatter Plot Enhancement: The Y-axis now plots 'Items Corrected' while the marker 'Size' dynamically scales with the 'Retention Score'. This creates a three-dimensional view of customer value: Spending (Position), Activity (Height), and Loyalty (Size).
- 2. Predictive KPI Cards: A centralized 'Average Retention Score' card allows management to monitor the health of specific segments (e.g., specific cities) in real-time as slicers are applied.

5. Strategic Decision-Support Use Cases

Operational Focus	Targeted Audience	Research-Based Action
Revenue Risk Mitigation	At-Risk Customers	Identify high-spenders with Scores < 1.0 for VIP recovery calls.
Growth & Expansion	Market Analysts	Compare Retention Scores across Decoded Cities to prioritize logistics hubs.
Loyalty Optimization	Marketing Team	Incentivize Silver members with scores > 2.0 to upgrade to Gold tiers.

6. Conclusion

By successfully implementing Milestone 6, the system demonstrates advanced proficiency in Data Science methodologies. The transition from binary data processing to predictive loyalty modeling provides a scalable framework for e-commerce growth. The system successfully meets all criteria for integrated research and technical innovation.